SENG3030 – INTERNET OF THINGS

ASSIGNMENT 4

Multi-Device IoT Prototype over MQTT

Architecture and Communications Report

TEAM

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1. INTRODUCTION

Our team of four built a multi-device IoT prototype in which three M5StickC Plus devices [1] and one web page exchange data only through the course MQTT broker [2]. The four components are:

- **A · Data acquisition device:** reads all lab sensors (SHT30, QMP6988, IMU, battery, microphone) and publishes the data.
- **B · Data display device:** shows the data from A on its screen, one sensor per page, and plays audio clips from A.
- **C · Control device:** changes the page on B, turns all data from A on or off, and asks A for an audio clip.
- **D · Client UI:** a web page that shows all data, the state of each device and the audio clips.

2. ARCHITECTURE

Figure 1 shows the overall communications. The broker (EMQX) lets each account publish only under `SENG3030/Wednesday/<own username>/`, but subscribe to any topic under `SENG3030/` [3]. Each device therefore logs in with a different team member's account and publishes in its own namespace; a device receives commands by subscribing to the sender's namespace. The devices connect with MQTT over TLS on port 8883 using PubSubClient [4]. A browser cannot open a raw TCP connection, so D uses MQTT over secure WebSocket on port 8084 [5] [6].

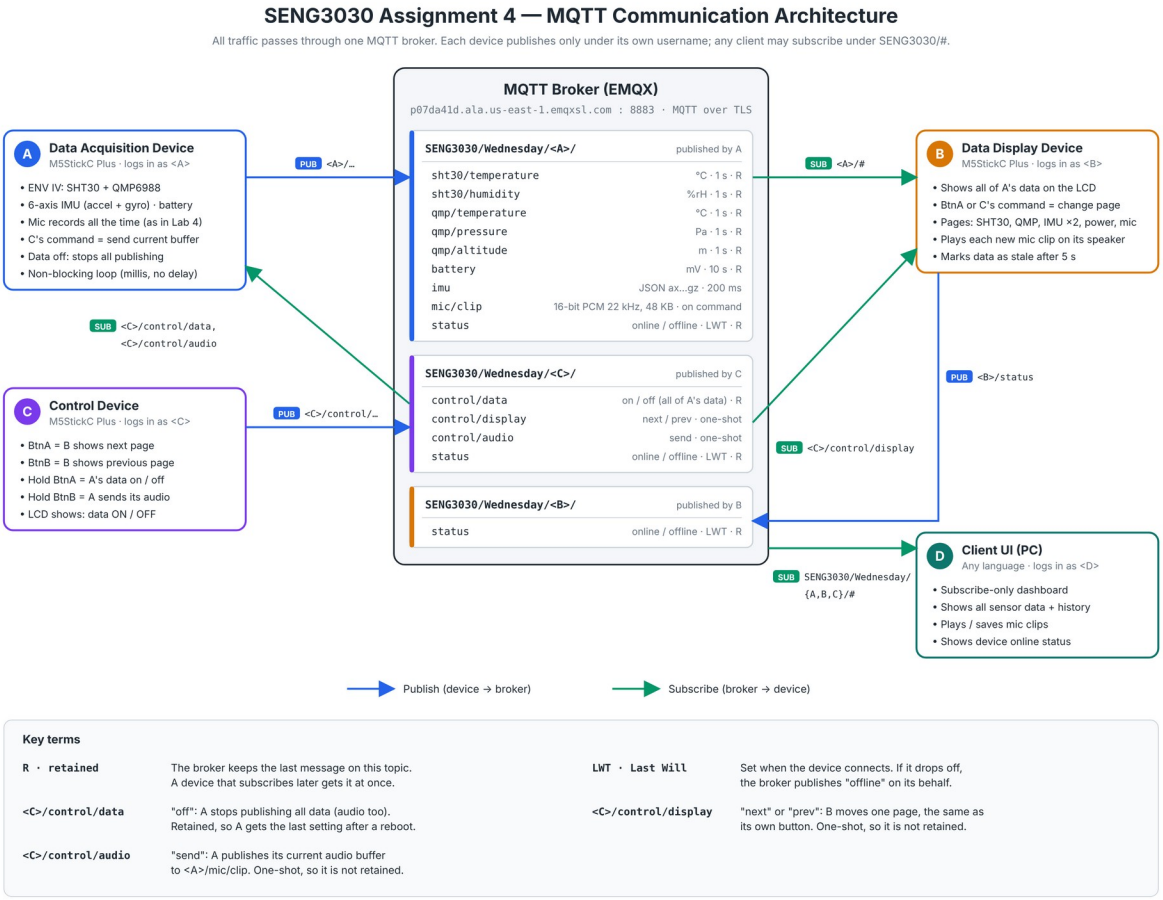


Figure 1: Block diagram of the overall communications

Table 1: Components and MQTT accounts

Component	MQTT account	Publishes	Subscribes to
A · Data acquisition	jbejaranoherrer7056	sensor data, mic/clip, status	<C>/control/data, <C>/control/audio
B · Data display	cgomezguerrero5580	status	<A>/#, <C>/control/display
C · Control	dzou1734	control/data, control/display, control/audio, status	(none)
D · Client UI (browser)	atu8211	(none)	<A>/#, <B>/#, <C>/#

<A>, <B>, <C> = SENG3030/Wednesday/<username of A, B or C>/

3. MESSAGES

All messages use QoS 0. Numbers are sent as text. **Retained** means the broker keeps the last value and gives it to a new subscriber at once [2]. Each device sets `status = offline` as its **Last Will**, so the broker publishes it if the device drops off [2].

Table 2: Messages published by A (prefix <A>/), received by B and D

Topic	Payload (example)	Unit	Rate	Retained
`sht30/temperature`	`23.45`	°C	1 s	Yes
`sht30/humidity`	`41.20`	%RH	1 s	Yes
`qmp/temperature`	`23.80`	°C	1 s	Yes
`qmp/pressure`	`101325.00`	Pa	1 s	Yes
`qmp/altitude`	`312.40`	m	1 s	Yes
`battery`	`4012`	mV	10 s	Yes
`imu`	`{"ax":0.012,"ay":-0.981,"az":0.105, "gx":1.25,"gy":0.30,"gz":-0.48}`	g; °/s	200 ms	No
`mic/clip`	`48 000 bytes: 16-bit PCM, mono, 22 050 Hz, about 1.1 s`	–	When C asks	No
`status`	`online / offline`	–	Connect / drop	Yes

Table 3: Messages published by C (prefix <C>/)

Topic	Payload	Button on C	Receiver: effect	Retained
`control/display`	`next / prev`	Click BtnA / BtnB	B: next / previous page	No
`control/data`	`on / off`	Hold BtnA	A: start / stop all publishing	Yes
`control/audio`	`send`	Hold BtnB	A: publish its audio buffer on mic/clip	No
`status`	`online / offline`	–	D: device status	Yes

Table 4: Message published by B (prefix <B>/)

Topic	Payload	Receiver	Retained
`status`	`online / offline`	D: device status	Yes

4. COMPONENTS

A reads each sensor on its own `millis()` timer and never calls `delay()`. WiFi and MQTT reconnect automatically. As in Lab 4 [7], the microphone records all the time into a ring buffer; on `send` A publishes the buffer. The clip uses 22 050 Hz instead of 44 100 Hz because PubSubClient cannot handle a message of 64 KB or more [4].

B has six pages: SHT30, QMP6988, accelerometer, gyroscope, battery and microphone. The front button or a command from C changes the page. A value turns red (STALE) when it has not been updated for 5 s (25 s for the battery). New audio clips play on the speaker.

C sends one command per button action (Table 3). A beep confirms that the command was sent, and the screen shows whether A's data is ON or OFF.

D is one HTML file that uses MQTT.js [8] and Chart.js [9]. It shows the online state of A, B and C, every value from A with its age, live charts, the audio clips (play or save as .wav) and the latest MQTT messages.

REFERENCES

[1] M5Stack, "M5StickC PLUS," M5Stack Docs. [Online]. Available: https://docs.m5stack.com/en/core/m5stickc_plus

[2] OASIS, "MQTT Version 3.1.1," OASIS Standard, Oct. 2014. [Online]. Available: http://docs.oasis-open.org/mqtt/mqtt/v3.1.1/os/mqtt-v3.1.1-os.html

[3] Conestoga College, "SENG3030 Lab 3 – Sensors and Such," course handout, 2026.

[4] N. O'Leary, "Arduino Client for MQTT (PubSubClient)," GitHub. [Online]. Available: https://github.com/knolleary/pubsubclient

[5] EMQ Technologies, "Create a Serverless Deployment," EMQX Cloud Docs. [Online]. Available: https://docs.emqx.com/en/cloud/latest/create/serverless.html

[6] EMQ Technologies, "How to Use MQTT over WebSocket: A Beginner's Guide," EMQX Blog. [Online]. Available: https://www.emqx.com/en/blog/connect-to-mqtt-broker-with-websocket

[7] Conestoga College, "SENG3030 Lab 4 – Microphone and Speaker," course handout, 2026.

[8] MQTT.js contributors, "MQTT.js," GitHub. [Online]. Available: https://github.com/mqttjs/MQTT.js

[9] Chart.js contributors, "Chart.js Documentation." [Online]. Available: https://www.chartjs.org/docs/latest/

EVALUATION FORM

All evaluation criteria are judged on a scale from zero to five.

Document		**Self Evaluation**	**Score**
	Completeness, Format, Content	/ 5	/ 5
Demonstration		**Self Evaluation**	**Score**
	Demonstration and Readiness	/ 5	/ 5
Functionality		**Self Evaluation**	**Score**
	Data Acquisition	/ 5	/ 5
	Data Display	/ 5	/ 5
Groups of 3	Remote Control	/ 5	/ 5
Groups of 4	Client Display	/ 5	/ 5
	Weighted Functionality Grade (scaled to 10)	/ 10	/ 10
Assignment Total			/ 20